From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA TEST SERIES

FIITJEE

JEE(Advanced)-2014

ANSWERS, HINTS & SOLUTIONS

CRT – III
(Paper-2)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	C	B	A
2.	D	B	C
3.	A	B	C
4.	A	A	C
5.	D	B	B
6.	C	B	B
7.	C	D	B
8.	B	C	D
9.	A, B, C, D	A, B	A, B, C
10.	A, C	A, C	A, B, C, D
11.	A, D	A, B, C	A, B, C, D
12.	A, B, C	A, D	B, C
1.	(A) → (q) (B) → (q) (C) → (r) (D) → (s)	(A) → (p, s) (B) → (p, r, s, t) (C) → (p, r, s) (D) → (r)	(A) → (t) (B) → (r) (C) → (q) (D) → (s)
2.	(A) → (q), (s) (B) → (p), (s) (C) → (p), (s) (D)→ (p), (r)	(A) → (q, s) (B) → (q, r) (C) → (p) (D) → (s)	(A) → (s) (B) → (t) (C) → (r) (D) → (s)
1.	2	9	4
2.	2	3	7
3.	5	6	4
4.	5	7	5
5.	2	4	4
6.	1	5	9

Physics

PART – I

1. In first case, $3 = k(2 - \ell_0)$, If ℓ_0 is natural length of the spring
In second case $4 = k(\ell_1 - \ell_0)$, ℓ_1 is the length of the spring in second case
By solving the equations we have $\ell_0 = 8 - 3\ell_1$
but ℓ_0 is always greater than zero.
$\Rightarrow \quad 8 - 3\ell_1 > 0$
$\Rightarrow \quad \ell_1 < \frac{8}{3}$ so it cannot be 2.7 m.

2. $U = 2x + 4y$
$F = -\frac{\partial U}{\partial x}\hat{i} - \frac{\partial U}{\partial y}\hat{j} - \frac{\partial U}{\partial z}\hat{k} = \left(-2\hat{i} - 4\hat{j}\right)N$
At t = 0, at (4, 4), P = 24 J
Kinetic energy K = 0
Total energy, E = P + k + 24J and acceleration $\vec{a} = \left(-\hat{i} - 2\hat{j}\right)m/s^2$
To cross x-axis, along y, $4 = \frac{1}{2}a_y t_1^2 = \frac{1}{2}2t_1^2 \qquad \Rightarrow t_1 = 2\text{sec}$
To cross y-axis, along y, $4 = \frac{1}{2}a_x t_2^2 = \frac{1}{2}.1.t_2^2 \qquad \Rightarrow t_2 = 2\sqrt{2}$ sec
Also, Crossing the x-axis $\Rightarrow$ co-ordinates are $\left(4 - \frac{1}{2}a_x t_1^2, 0\right) = (2,0)$
$\therefore$ P = 4J $\Rightarrow$ K = E – P = 20 J.
Crossing the y-axis $\Rightarrow$ co-ordinates are $\left(0, 4 - \frac{1}{2}a_y t_2^2\right) = (0,-4)$
$\therefore$ P = –16J, K = E – P = 40 J

3. If at $t = t_0$ container is empty then the rate by which water comes out is $\frac{m}{t_0}$. At any t, total mass
$m_1 = m + m - \frac{mt}{t_0}$
$\therefore \quad F - m_1 g = m_1 a = m_1 \frac{dv}{dt}$
$\Rightarrow \quad \int_0^v dv = \int_0^{t_0} \frac{F}{m_1}dt - \int_0^{t_0} g\,dt$
$\Rightarrow \quad v = \int_0^{t_0} \frac{F}{2m - \frac{mt}{t_0}} - gt_0 = \frac{t_0 F}{m}.\ell n2 - gt_0$
$\Rightarrow mv = t_0[F\ell n2 - mg]$
$\Rightarrow \quad t_0 = \frac{mv}{F(\ell n2) - mg}$ correct coice is (A)

4. $V = \frac{K}{\sqrt{T}}$, $T = \frac{PV}{nR}$ is put then PV^3 = constant (say k_1)

$\therefore \Delta W = PdV = \int \frac{k_1}{v^3} dv = \frac{-nR\Delta T}{2} = \frac{-5R\Delta T}{2}$

$\Delta U = nCv\Delta T = (n_1Cv_1 + n_2Cv_2)\Delta T = \left(3x\frac{3R}{2} + 2x\frac{5R}{2}\right)\Delta T = \frac{19}{2}R\Delta T$

$\Delta Q = \Delta U + \Delta W = \frac{14R\ \Delta T}{2} = 7R\Delta T$

$C_{(process)} = \frac{\Delta Q}{n\Delta T} = \frac{7R\Delta T}{5\Delta T} = \frac{7R}{5}$

5. Total flux coming out of the sphere is $\frac{Q}{\epsilon_0}$ and conductivity at that point, $\sigma = \frac{1}{\rho}$

The relation current density, $J = \sigma E$

$\therefore$ rate of change of charge, $-\frac{dQ}{dt} = \vec{J}.\vec{A} = \sigma\vec{E}\cdot\vec{A}$ = σ[Total flux]

$-\frac{dQ}{dt} = \sigma.\frac{Q}{\epsilon_0} = \frac{Q}{\rho\epsilon_0}$

Hence, $-\int_{Q}^{Q/4}\frac{dQ}{Q} = \int_0^{t_1}\frac{dt}{\rho\epsilon_0} \Rightarrow \ell n4 = \frac{t_1}{\rho\ \epsilon_0} \Rightarrow t_1 = \rho\epsilon_0\ \ell n4 = 2\rho\ \epsilon_0\ (\ell n2)$

6. If the loop is completed by wire of length 2R along y-axis i.e. the diametre of the semicircle then $\sum\vec{B}.\vec{d\ell} = \mu_0 I$ [Ampere's law]. And due to the diameter wire only,

$\left|\vec{B}\right|$ at all point on the circle $3x^2 + 3z^2 = R^2$, is $\frac{\mu_0 I}{4\pi r}\left[\sin\theta_1 + \sin\theta_2\right] = \frac{\mu_0 I}{4\pi\frac{R}{\sqrt{3}}}\left[\sin 60^o + \sin 60^o\right] = \frac{3\mu_0 I}{4\pi R}$

$\therefore \sum\vec{B}.\vec{d\ell}_{\text{(due to diameter wire only)}} = \frac{3\mu_0 I}{4\pi R}\times 2\pi\frac{R}{\sqrt{3}} = \frac{\sqrt{3}}{2}\mu_0 I$

Hence, $\sum\vec{B}.\vec{d\ell}$ due to semicircle = $\mu_0 I - \frac{\sqrt{3}\mu_0 I}{2} = \frac{\left(2-\sqrt{3}\right)\mu_0 I}{2}$

7. Intesnity, $I = 4I_0\cos^2\frac{\varphi}{2}$

$\therefore I_{max} = 4I_0$, For I to be $2I_0$.

Phase diff. $\varphi = (2n+1)\frac{\pi}{2}$, n = 0, 1, 2, 3.....

$\therefore$ Path diff. = $\frac{\varphi}{2\frac{\pi}{\lambda}} = \frac{\lambda}{4}(2n+1)$; $\therefore\ \frac{hd}{D} = \frac{\lambda}{4}(2n+1)$

$\Rightarrow \lambda = \frac{4hd}{(2n+1)D} = \frac{4hd}{(\text{odd integer})xD}$

8. $P = at + \frac{b+t^2}{c} + dt^3$

$[P] = [m.v] = [MLT^{-1}]$

$[a] = \frac{[P]}{[t]} = MLT^{-2}$

$[b] = [t^2] = M^oL^oT^2$

$\left[\frac{t^2}{c}\right] = [P] \Rightarrow [c] = \frac{[t^2]}{[P]} = M^{-1}L^{-1}T^3$

$[d] = \frac{[P]}{[t^3]} = MLT^{-4}$

11. To have max. M.E. on ground spring should be detached at a point where spring have minimum energy.

12. If $I = \frac{mR^2}{K}$, $\mu_{min} = \frac{\tan\theta}{1+K}$

K is minimum for cylindrical shell $\Rightarrow \mu_{min} = \frac{\tan\theta}{1+2} = \frac{1}{2}\tan\theta$ is maximum

$\therefore$ If shell can roll then other will also roll.
Force of frictional is static hence work done by force will zero w.r.t. ground.

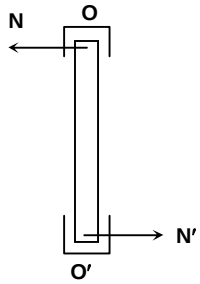
SECTION–B

1. Length of wire (Young's modulus Y_2). Cos 60^0 = length of wire (Young's modulus Y_1)
By symmetry and small deformation
Elongation in wire (Young's modulus Y_2) = Elongation in wire (Young's modulus Y_1).cos 60^0
4*Strain in wire (Young's modulus Y_2) = strain in wire (Young's modulus Y_1)

2. Degree of freedom f = 3
$\Delta U = ½ fnR(T_2 – T_1) = ½ f(P_2V_2 – P_1V_1) = ½ fP_1V_1\{(V_2/V_1)^{1-n} – 1\}$; > 0 for n < 1 and < 0 for n > 1
$\Delta W = (P_1V_1 – P_2V_2)/(n-1) = P_1V_1\{1 - (V_2/V_1)^{1-n}\}/(n-1)$;
$\Delta Q = \Delta U + \Delta W = P_1V_1\{(V_2/V_1)^{1-n} - 1\}\{ ½ f - 1/(n-1)\}$;

SECTION–C

1. About O



$N'\ell + \tau_{centrifugal} + \left(-mg\frac{\ell_o}{2}\right) = 0$

About O′

$N\ell + \tau_{centrifugal} + \left(-mg\frac{\ell_o}{2}\right) = 0$

$\tau_{centrifugal} = \int_0^{\ell_o} \frac{m\omega^2\ell}{2\ell_0} x dx = \frac{1}{4}m\omega^2\ell\ell_o$ (anticlockwise)

$\therefore \quad N'\ell + \frac{1}{4}m\omega^2\ell\ell_o - mg\frac{\ell_o}{2} = 0$

and $N\ell - \frac{1}{4}m\omega^2\ell\ell_o - mg\frac{\ell_o}{2} = 0$

When N′ = 0, $\frac{1}{4}m\omega^2\ell\ell_o - mg\frac{\ell_o}{2} = 0$

$\Rightarrow \quad \omega = \sqrt{\frac{2g}{\ell}}$

2. $2T\cos\theta\left(\frac{1}{r_1}-\frac{1}{r_2}\right)=\rho gh$

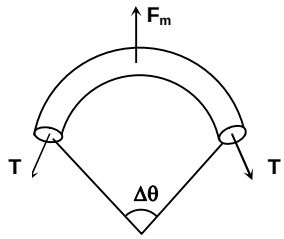
3. $\nu=\frac{C_p}{C_v}\quad\left(C_p=\frac{\Delta Q}{\Delta T},\ C_v=\frac{\Delta U}{\Delta T}\right)$

4. $2T\sin\frac{\Delta\theta}{2}=Bi\,(\Delta\theta R)$

$\Rightarrow T=IRB$

also, $T=\pi r^2\sigma$

$\therefore B=\frac{\pi r^2\sigma}{IR}$



Chemistry

PART – II

SECTION – A

1. $HCl + NaOH \longrightarrow NaCl + H_2O$

$H^+ + OH^- \longrightarrow H_2O$; x kJ

$CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$

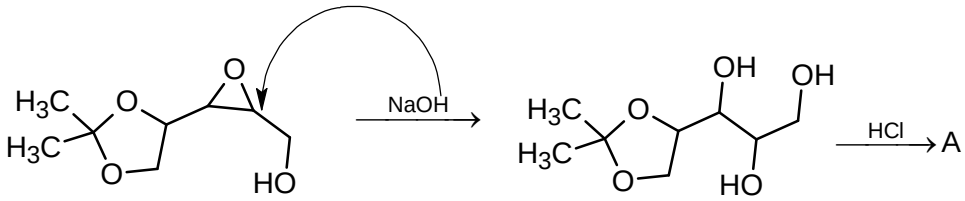
$CH_3COOH + OH^- \longrightarrow CH_3COO^- + H_2O$; y kJ

Required equation:

$CH_3COOH \longrightarrow CH_3COO^- + H^+$

$\therefore \Delta H = (y - x)kJ$

2. Because it has one d and one ℓ centre.

4. 

5. Solubility of Ag_2CrO_4 $S_1 = \sqrt[3]{\frac{K_{sp}}{4}}$

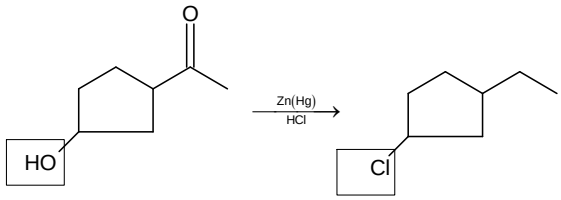
$= 6.5 \times 10^{-5}$ M

Solubility of $AgBrS_2 = \sqrt{K_{sp}}$

$= 7.07 \times 10^{-7}$ M

Ratio = $\frac{S_1}{S_2} = \frac{6.5 \times 10^{-5} M}{7.07 \times 10^{-7}}$

= 91.9

6. (B) 

7. 12 coordination (f.c.c.) **:** 8 coordination (b.c.c.)

$\frac{4 \times M_x}{N_{av} \times a^3}$ **:** $\frac{2 \times M_x}{N_{av} \times a^3}$ (Ratio of density)

For f.c.c. lattice, $\Rightarrow a = \frac{4r}{\sqrt{2}}$

For b.c.c. lattice, $4r = \sqrt{3}a \Rightarrow a = \frac{4r}{\sqrt{3}}$

Ratio of density = $\dfrac{\frac{4}{N_{av}}\times M_x}{\left(\frac{4r}{\sqrt{2}}\right)^3} : \dfrac{\frac{2}{N_{av}}\times M_x}{\left(\frac{4r}{\sqrt{3}}\right)^3}$

$= 2 \times \left(\sqrt{2}\right)^3 : \left(\sqrt{3}\right)^3$

9. Because magnitude of $KE = \dfrac{ze^2}{2r}$, $PE = \dfrac{-ze^2}{r}$ and $TE = \dfrac{-ze^2}{2r}$

10. According to Le Chatelier's principle.

11. A is present at corners and face centres and B in the tetrahedral voids.
So, along one axis two face centred atoms will be and if they are removed then A are present in the corner and centre of four faces.
So, total no. of atoms of A = 1 + 2 = 3
total no. of atoms of B = 4
Formula is A_3B_4

12. Ba^{2+} ions produce yellow ppt. with CrO_4^{2-}, which is not soluble in CH_3COOH.
$BaCO_3$ dissolves in CH_3COOH and produces Ba^{2+} ion which gives yellow ppt. with CrO_4^{2-} which is insoluble in CH_3COOH.
$SrCO_3$ is soluble in CH_3COOH and produces Sr^{2+} ion which does not give yellow ppt. of $SrCrO_4$ in CH_3COOH.

SECTION – C

1. $\log\left(\dfrac{10\times10^3}{4.5\times10^3}\right) = \dfrac{58\times10^3}{2.303\times8.314}\left(\dfrac{T_2-274}{274T_2}\right)$
$\Rightarrow T_2 = 283\ K = 10^oC$
$\therefore$ Rise in temperature = $10 - 1 = 9^oC$

2. CH_3COCHO, $OHCCH_2CH_2CHO$

4. $\left[H^+\right] = \dfrac{10\times10^{-5}}{10{,}000} = 10^{-8}$
$\left[H^+\right]_{total} = 10^{-8}\left(\text{from } H_2SO_4\right) + 10^{-7}\left(\text{from water}\right)$
$= 1.1 \times 10^{-7}$
$pH = 7 - \log 1.1 = 6.94 \approx 7$

5. $T_c = \dfrac{8a}{27Rb}, V_c = 3b, P_c = \dfrac{a}{27b^2}$
$Z_c = \dfrac{P_cV_c}{RT_c} = \dfrac{3}{8} = \dfrac{1.5}{4}$

6. Black sulphides are Hgs, PbS, CuS, NiS, CoS.

Mathematics PART – III

SECTION – A

1. $I = \int_{-\infty}^{0} e^x dx + \int_{0}^{\ln 2} (e^x - 1)dx + \int_{\ln 2}^{\ln 3} (e^x - 2)dx$

On solving = $3 + \ln 2 - 2 \ln 3$

2. $z = -4(i-1) \pm \sqrt{-63-16i}$

$z = -4(i-1) \pm (1 - 8i)$

$z = 5 - 12i, 3 + 4i$

Hence area $(\Delta oz_1z_2) = \frac{1}{2}\begin{vmatrix} 0 & 0 & 1 \\ 5 & -12 & 1 \\ 3 & 4 & 1 \end{vmatrix} = \frac{1}{2} \times 56 = 28$

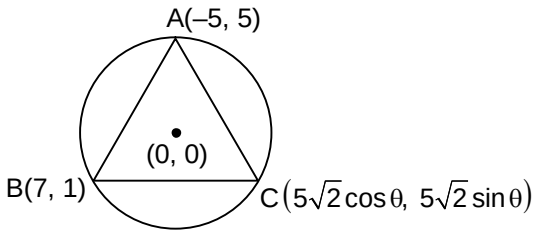
3. $\frac{h}{3} = \frac{2 + 5\sqrt{2}\cos\theta}{3}$

and $\frac{k}{3} = \frac{4 + 5\sqrt{2}\sin\theta}{3}$

Orthocenter 2 — 1 Circumcentre

H(h, k) — G — (0, 0)

$\left(\frac{-5 + 7 + 5\sqrt{2}\cos\theta}{3}, \frac{5 - 1 + 5\sqrt{2}\sin\theta}{3}\right)$

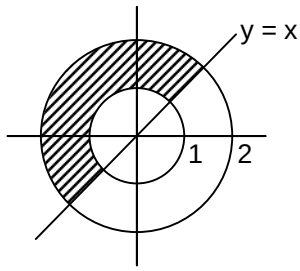


On eliminating θ, we get $(x-2)^2 + (y-4)^2 = 50$

$x^2 + y^2 - 4x - 8y - 30 = 0$

4. $k = \frac{\pi(2^2 - 1^2)}{2} = \frac{3\pi}{2}$

$\int_0^{\pi} \{\sin x\} dx = \int_0^{\pi} \sin x dx = 2$



5. $h^2 + 16 = (h-8)^2$

$h^2 + 16 = h^2 - 16h + 64$

$h = 3$

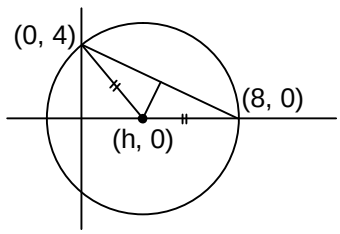
So centre is (3, 0) and r = 5

equation of circle $(x-3)^2 + y^2 = 25$

Let $C(3 + 5\cos\theta, 5\sin\theta)$

$\Delta = \frac{1}{2}\begin{vmatrix} 0 & 4 & 1 \\ 8 & 0 & 1 \\ 3 + 5\cos\theta & 5\sin\theta & 1 \end{vmatrix} = .10|(2\sin\theta + \cos\theta - 1)|$

$A_{max} = 10(\sqrt{5} + 1)$



6. Slope of given line = $-\frac{3}{2}$

First of all we try to locate points on the curve at which tangents are parallel to given line
On differentiating $3x^2 - 4y^2 = 72$ w.r.t. x

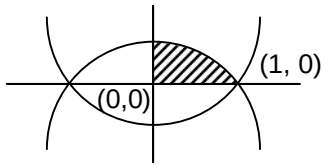
$\frac{dy}{dx} = \frac{3x}{4y} = -\frac{3}{2}$ (Given) $\frac{x}{y} = -2$

Put $x = -2y$ in $3x^2 - 4y^2 = 72$

We get (–6, 3) and (6, –3) and their distances from line are $\frac{11}{\sqrt{13}}$ and $\frac{13}{\sqrt{13}}$ respectively

So (–6, 3) is nearest
So $x_0 = -6 \Rightarrow y_0 = 3$ and $(x_0 + y_0) = -3$

7. $y^2 = (x^2 - 1)^2$
So $y = x^2 - 1$ or $y = 1 - x^2$

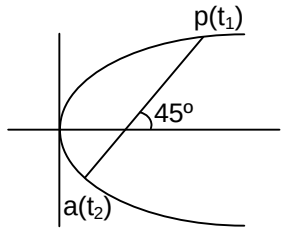


So required area = $4\int_0^1 (1-x^2)dx = 4\left[x - \frac{x^3}{3}\right]_0^1 = \frac{8}{3}$

8. Given limit = $0 + (2^2 - 1) + (3^2 - 1) + ... + (10^2 - 1) = \sum_{n=2}^{10}(n^2 - 1)$

$= \frac{10 \times 11 \times 21}{6} - 1 - 9 = 375$

9. Circle on PQ as diameter



$(x - t_1^2)(x - t_2^2) + (y - 2t_1)(y - 2t_2) = 0$

$x^2 + y^2 - (t_1^2 + t_2^2)x - 2(t_1 + t_2)y + t_1^2 t_2^2 + 4t_1 t_2 = 0$

Put $\frac{t_1 + t_2}{2} = 1$ and $t_1 t_2 = -1$

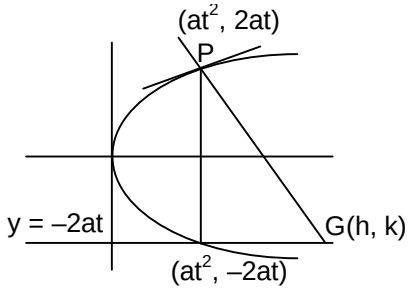
Hence equation of circle become
$x^2 + y^2 - 6x - 4y - 3 = 0$ ..... (1)
With centre (3, 2) and radius 4
Equation (1) is not orthogonal to $x^2 + y^2 + 2x - 6y + 3 = 0$

10. Three planes meet at two points it means they have infinite many solution, so

$\begin{vmatrix} 2 & 1 & 1 \\ 1 & -1 & 1 \\ \alpha & -1 & 3 \end{vmatrix} = 0$ on solving $\alpha = 4$

$P_1 : 2x + y + z = 1$
$P_2 : x - y + z = 2$
$P_3 : 4x - y + 3z = 5$
P on XOY plane $\equiv$ (1, –1, 0) (Which can be obtained by putting z = 0 in any two equations)
Q on YOZ plane $\equiv \left(0, -\frac{1}{2}, \frac{3}{2}\right)$ (By putting x = 0 in any two equations)

11. $N : y + tx = 2at + at^3$
$L : y = -2at$
On solving together $-2at + tx = 2at + at^3$
$x = at^2 + 4a$
Hence $h = at^2 + 4a$ ..... (1)
$k = -2at$ ..... (2)
So $y^2 = 4a(x - 4a)$ (eliminating t)
Now verify alternatives



12. $4x^2 + 2x - 1 = 0$

$$x = \frac{-2 \pm \sqrt{4+16}}{8}$$

$$x = \frac{-2 \pm 2\sqrt{5}}{8} = \frac{-1 \pm \sqrt{5}}{4}$$

If $\alpha = \frac{\sqrt{5}-1}{4} = \sin 18^\circ$

$$\beta = -\left[\frac{\sqrt{5}+1}{4}\right] = -\cos 36^\circ$$

$\beta = -\cos 36^\circ = -\sin[90^\circ - 36^\circ] = -\sin 54^\circ = 4\sin^3 18^\circ - 3\sin 18^\circ = 4\alpha^3 - 3\alpha$
$\beta = -\cos 36^\circ = 2\cos^2 18^\circ - 1 = 2\alpha^2 - 1$

SECTION – B

1. (A) $\lim\limits_{x \to 3} f(x) = 4$
(B) $x^2 + 6x + 9 + y^2 = 9 + 4$
$\therefore (x + 3)^2 + y^2 = 9 + 4$
$\therefore (x + 3)^2 = 9, y^2 = 4$ or $(x + 3)^2 = 4, y^2 = 9$
(C) Draw graph and check
(D) Let $\alpha = 2 - \sqrt{95}$ and β be other root
As $\alpha + \beta = -a$ (Given)
$\beta = -a - \alpha = -a - 2 + \sqrt{95}$
Also $\alpha\beta = b$
$\left(2 - \sqrt{95}\right)\left(-a - 2 + \sqrt{95}\right) = b$
But $b < 0$ (Given)
$\Rightarrow \left(-a - 2 + \sqrt{95}\right) > 0$
So $a < \sqrt{95} - 2$
Hence max of a = 7

2. (A) $\cos x + \cos 3x = \cos 2x$
$2\cos 2x \cos x - \cos 2x = 0$
$\cos 2x (2\cos x - 1) = 0$
$\cos x = \frac{1}{2} \therefore x = \frac{\pi}{3}$ or $\frac{5\pi}{3}$ or $\cos 2x = 0$
$\Rightarrow 2x = \frac{\pi}{2}$ or $\frac{3\pi}{2} \Rightarrow x = \frac{\pi}{4}, \frac{3\pi}{4}$
Roots are $\left\{\frac{\pi}{3}, \frac{5\pi}{3}\right\}$
$\Rightarrow$ 2 roots

(B) $f'(x) = 2x - \frac{a}{x^2}$

$f'(2) = 4 - \frac{a}{4} = 0$

$\Rightarrow$ a = 16 so a_1 = 16

Also f″(x) = $2 + \frac{2a}{x^3}$

Also f″(1) = 2 + 2a = 0

a = –1 so a_2 = –1

Hence $\frac{a_1 + a_2}{3} = \frac{15}{3} = 5$

(C) Check by graph

(D) We have $\lim_{x\to 0} \frac{\int_0^x \frac{t^2}{\sqrt{a+t}}dt}{\left(\frac{x - \sin x}{x^3}\right)x^3} = 1$

6. $\lim_{x\to 0} \frac{x^2}{\sqrt{a+x}} \times \frac{1}{3x^2} = 1 = \frac{2}{\sqrt{a}} = 1$ so a = 4

SECTION – C

1. $^9C_3 \times {}^3C_2 \times \frac{5!}{2!2!1!}$

2. $f(x) = (1+x)^{20}\left[1 + \frac{x}{1+x} + \left(\frac{x}{1+x}\right)^2 + ... + \left(\frac{x}{1+x}\right)^{18}\right]$

$(1+x)^{21}\left[1 - \left(\frac{x}{1+x}\right)^{19}\right] = (1+x)^{21} - (1+x)^2 x^{19}$

Coefficient of x^{18} in f(x) = Coefficient of x^{18} in $(1 + x)^{21}$ = $^{21}C_{18}$ = 1330

3. Let 2^x = t (for x < 0, t < 1 and for x > 0, t > 1)
Now g(t) = t^2 + 2(1 – b)t + b = 0
Unity must be between the roots of this equation g(1) < 0
1 + 2(1 – b) + b < 0 $\Rightarrow$ b > 3
Possible values of b in [1, 20] are 4, 5, 6, ... 19, 20
So p = 204 = 4

4. Area of $\Delta ABC = \frac{1}{2}|\vec{a}\times\vec{b} + \vec{b}\times\vec{c} + \vec{c}\times\vec{a}|$

Now given $2\vec{a} + 3\vec{b} + 6\vec{c} = 0$

Cross with $\vec{a}$, $3\vec{a}\times\vec{b} + 6\vec{a}\times\vec{c} = 0$

or $\vec{a}\times\vec{b} = 2(\vec{c}\times\vec{a})$ ..... (1)

Again cross with $\vec{b}$

$2\vec{a}\times\vec{b} + 6\vec{c}\times\vec{b} = 0$

or $\vec{a}\times\vec{b} = 3(\vec{b}\times\vec{c})$

Area of $\Delta OAB = \frac{1}{2}|\vec{a}\times\vec{b}|$

$$\frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta AOB} = \frac{\frac{1}{2}|\vec{a}\times\vec{b}|\left(1+\frac{1}{2}+\frac{1}{3}\right)}{\frac{1}{2}|\vec{a}\times\vec{b}|} = \frac{11}{6}$$

So m = 11, n = 6 $\Rightarrow$ (m – n) = 5

5. $x^2 + (2 - \tan\theta)x - (1 + \tan\theta) = 0$

$\alpha + \beta = \tan\theta - 2$ ..... (1)

$\alpha\beta = -\tan\theta - 1$ ..... (2)

From equation (1) + (2)

$\alpha + \beta + \alpha\beta = -3$

$(\alpha + 1)(\beta + 1) = -2$

Hence either $\alpha + 1 = -2$ and $\beta + 1 = 1$

then $\alpha = -3$, $\beta = 0$

or $\alpha + 1 = -1$ and $\beta + 1 = 2$

$\alpha = -2$ and $\beta = 1$

If $\alpha = -3$, $\beta = 0$ then $\tan\theta = -1$

$\Rightarrow \alpha = \frac{3\pi}{4}, \frac{7\pi}{4}$

If $\alpha = -2$, $\beta = 1$ then $\tan\theta = 1$

$\Rightarrow \theta = \frac{\pi}{4}, \frac{5\pi}{4}$

Sum = $\frac{3\pi}{4} + \frac{7\pi}{4} + \frac{\pi}{4} + \frac{5\pi}{4} = 4\pi = 4$

6. $$\frac{p}{q} = 1 - \frac{{}^{9}C_8 \cdot {}^{1}C_1}{{}^{10}C_9} \times \frac{{}^{18}C_9}{{}^{19}C_9} = 1 - \frac{9}{19} = \frac{10}{19}$$

So q – p = 19 – 10 = 9